7th, KM Haridwar, National Highway Vardhmanpuram,
Roorkee, Rehmadpur, Uttarakhand 247667

Lab Sheet-03

Python Programming Lab – String and List Operations

MCA I Semester (Session 2026-2027)

Objectives:

- To develop problem-solving skills using string manipulation techniques in Python.
- To implement list operations for handling and processing collections of data.

Requirements:

- Python 3.x installed on the system (IDLE, PyCharm, VS Code, or any preferred IDE).
- Basic knowledge of **strings, lists, loops, and functions** in Python.
- Understanding of input/output operations and conditional statements.

Program List:

1. Write a Python program to check if a string is a palindrome.
2. Count the number of vowels and consonants in a string.
3. Replace all spaces in a string with underscores.
4. Extract all uppercase characters from a string.
5. Remove all duplicate characters from a string.
6. Find the frequency of each character in a string.
7. Reverse a string without using slicing.
8. Convert a string into title case without using built-in functions.
9. Count the number of words in a string.
10. Find the longest word in a string.

7th, KM Haridwar, National Highway Vardhmanpuram,
Roorkee, Rehmadpur, Uttarakhand 247667

11. Check if two strings are anagrams of each other.
12. Check if a substring exists in a given string.
13. Convert all vowels in a string to uppercase and consonants to lowercase.
14. Remove all punctuation from a string.
15. Find the first non-repeated character in a string.
16. Count the number of digits, alphabets, and special characters in a string.
17. Print all substrings of a string.
18. Sort the characters of a string alphabetically.
19. Find the most repeated word in a string.
20. Check if two strings are rotations of each other.
21. Find the largest and smallest elements in a list.
22. Sort a list of integers in ascending and descending order.
23. Remove duplicate elements from a list.
24. Reverse a list without using built-in functions.
25. Count the frequency of each element in a list.
26. Merge two lists into one.
27. Find the second largest number in a list.
28. Find common elements in two lists.
29. Find the difference between two lists.
30. Separate even and odd numbers from a list.

Instruction:

- Write each program in a separate .py file named according to the question (e.g., Q1_palindrome.py)
- Take inputs dynamically using the input () function.
- Provide comments to explain the logic.
- Display outputs clearly and in a user-friendly format.

Guideline:

- Use loops and conditional statements where applicable.
- Do not use built-in Python functions (like reversed(), set(), etc.) unless explicitly required.
- Follow proper indentation (4 spaces) and coding style.
- Validate inputs wherever necessary.
- Test your programs with different test cases.

Submission:

- Submit the source code files (.py) for each task along with a brief report documenting the implementation details and the results of the executions.



7th, KM Haridwar, National Highway Vardhmanpuram,
Roorkee, Rehmadpur, Uttarakhand 247667